

|                                                         |                                                       |
|---------------------------------------------------------|-------------------------------------------------------|
| Program : <b>Diploma in Instrumentation Engineering</b> |                                                       |
| Course Code : <b>3085</b>                               | Course Title: <b>Digital Circuits and Systems Lab</b> |
| Semester : <b>3</b>                                     | Credits: <b>1.5</b>                                   |
| Course Category: <b>Program Core</b>                    |                                                       |
| Periods per week: <b>3 (L: 0 T: 0 P: 3)</b>             | Periods per semester: <b>45</b>                       |

### Course Objectives:

- To familiarize the number system used in digital systems
- To familiarize combinational and sequential circuits.

### Course Prerequisites:

| Topic                                            | Course code | Course name                              | Semester |
|--------------------------------------------------|-------------|------------------------------------------|----------|
| Basic Electronics                                |             | Fundamentals of Electrical & Electronics | 2        |
| Number system, logic gates, flip flops, counters |             | Digital Circuits and Systems             | 3        |

### Course Outcomes:

On completion of the course, the student will be able to:

| COn | Description                                                                 | Duration (Hours) | Cognitive Level |
|-----|-----------------------------------------------------------------------------|------------------|-----------------|
| CO1 | Construct basic logic functions using TTL logic gates                       | 9                | Applying        |
| CO2 | Construct combinational logic circuits using logic gates                    | 10               | Applying        |
| CO3 | Construct flip flops and shift registers using gates and ICs                | 8                | Applying        |
| CO4 | Design and implement synchronous and asynchronous counters using Flip Flops | 12               | Applying        |
|     | Lab Test                                                                    | 6                |                 |

**CO – PO Mapping:**

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             | 3   |     |     |     |     |     |     |
| CO2             | 3   |     |     |     |     |     |     |
| CO3             | 3   |     |     |     |     |     |     |
| CO4             | 3   |     |     |     |     |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline:**

| Module Outcomes | Description                                                                                             | Duration (Hours) | Cognitive Level |
|-----------------|---------------------------------------------------------------------------------------------------------|------------------|-----------------|
| <b>CO1</b>      | <b>Construct basic logic functions using TTL logic gates</b>                                            |                  |                 |
| M1.01           | Verify the truth tables for all logic gates – NOT, OR, AND, NAND, NOR, XOR, XNOR using TTL Logic gates. | 3                | Understanding   |
| M1.02           | Design and implement logic circuits using universal gates                                               | 6                | Applying        |
| <b>CO2</b>      | <b>Construct combinational logic circuits using logic gates</b>                                         |                  |                 |
| M2.01           | Construct Half Adder, Full Adder, Half Subtractor and Full Subtractor using Logic gates                 | 5                | Applying        |
| M2.02           | Construct an alarm circuit using logic gates                                                            | 4                | Applying        |
| M2.03           | Construct 4 to 1 Multiplexer and 1 to 4 De-Multiplexer using Logic gates.                               | 1                | Applying        |
|                 | Lab Test – I                                                                                            | 3                | Applying        |
| <b>CO3</b>      | <b>Construct Flip Flops and Shift Registers</b>                                                         |                  |                 |
| M3.01           | Construct SR,D, JK and T Flip Flops using Logic gates                                                   | 4                | Applying        |
| M3.02           | Construct shift registers (left/right with mode control)                                                | 4                | Applying        |
| <b>CO4</b>      | <b>Design and implement synchronous and asynchronous counters using Flip Flop ICs.</b>                  |                  |                 |
| M4.01           | Design Asynchronous Decade Counter                                                                      | 3                | Applying        |
| M4.02           | Implement Synchronous Up-Down Counter                                                                   | 3                | Applying        |
| M4.03           | Implement Johnson counter using Flip Flops                                                              | 3                | Applying        |
| M4.04           | Implement Ring counter using Flip Flops                                                                 | 3                | Applying        |
|                 | Lab Test – II                                                                                           | 3                |                 |

**Text / Reference:**

| <b>T/R</b> | <b>Book Title/Author</b>                                                                                                                                       |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1         | Digital principles & Applications, Albert Paul Malvino & Donald P. Leach, McGraw Hill Education; Eighth edition ISBN: 978-9339203405                           |
| R2         | Digital Electronics, Roger L. Tokheim Macmillian, McGraw-Hill Education (ISE Editions);International 2 Revised edition ISBN: 978-0071167963 Macmillian         |
| R3         | Digital Electronics–an introduction to theory and practice, William H. Goth-mann, Prentice Hall India Learning Private Limited; 2 edition ISBN: 978-8120303485 |
| R4         | Fundamentals of Logic Design, Charles H. Roth Jr. Jaico Publishing House; First edition ISBN: 978-8172247744                                                   |
| R5         | Digital Electronics, R. Anand, Khanna Publications, New Delhi (Edition 2018) ISBN: 978-93-82609445                                                             |

**Online Resources:**

| <b>Sl. No</b> | <b>Website Link</b>                                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1             | <a href="https://en.wikibooks.org/wiki/Digital_Electronics">https://en.wikibooks.org/wiki/Digital_Electronics</a>                                                 |
| 2             | <a href="https://www.geeksforgeeks.org/digital-electronics-logic-design-tutorials/">https://www.geeksforgeeks.org/digital-electronics-logic-design-tutorials/</a> |
| 3             | <a href="https://www.allaboutcircuits.com/textbook/digital/">https://www.allaboutcircuits.com/textbook/digital/</a>                                               |